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Seismic Fault Detection Using Quantum Machine Learning

Gino Ananos and Saleh Al-Dossary, Saudi Aramco

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Abstract

This paper presents a groundbreaking application of Quantum Machine Learning (QML) in geophysics, specifically targeting seismic fault detection. By integrating quantum computing with machine learning, the study addresses critical limitations of traditional fault detection methods, such as the detection of subtle features, processing efficiency, and scalability. The primary objective is to enhance fault delineation accuracy and provide a robust framework suitable for large-scale seismic surveys and real-time applications.

The methodology involves encoding seismic data into quantum states to efficiently handle high-dimensional features. A hybrid quantum-classical model is developed, utilizing quantum-enhanced algorithms for fault detection. This model incorporates Quantum Long Short-Term Memory (QLSTM) networks, which leverage quantum parallelism and interference to enhance the processing capabilities of traditional LSTM networks. The QLSTM architecture employs quantum gates and memory cells to maintain and regulate the flow of quantum information, enabling more efficient and accurate processing of sequential data. The workflow encompasses quantum-based feature extraction, model training with seismic datasets, and validation using both synthetic and real-world examples. Comparative studies with traditional machine learning methods assess performance metrics, including accuracy, noise suppression, and computational efficiency. Quantum hardware simulators ensure the method's feasibility for practical applications.

Results demonstrate that QML, powered by QLSTM, significantly outperforms conventional methods in seismic fault detection, achieving higher accuracy, better resolution of subtle geological features, and reduced noise sensitivity. Comprehensive testing on synthetic and field datasets reveals QML's ability to capture complex fault patterns, particularly in geologically challenging settings. Additionally, QML's computational efficiency enables rapid analysis of large datasets, facilitating timely decision-making.

In conclusion, QML emerges as a superior approach for seismic fault detection, addressing longstanding challenges and offering novel opportunities for real-time seismic interpretation. By leveraging quantum computing, QML provides unprecedented insights into subsurface structures, leading to more accurate predictions and informed decisions in exploration and production operations.

This paper introduces a transformative approach to seismic fault detection, contributing significantly to existing literature and offering actionable insights for the petroleum industry, particularly in subsurface imaging and reservoir characterization.

Introduction

Seismic reflection surveying is a well-established method for obtaining subsurface information in the oil and gas exploration industry. By analyzing seismic data, experts can identify structural and stratigraphic geometric features, which, combined with other geophysical information, can indicate potential hydrocarbon accumulations. However, the interpretation of seismic data remains a time-consuming and human-intensive task, exacerbated by the ever-increasing volume of data and the tight deadlines demanded by the industry.

In response to the growing paradox between industry demand and increasing data volumes, deep learning technologies have emerged as powerful tools to assist geoscientists in seismic interpretation. Deep Neural Networks (DNNs) have demonstrated significant advancements in the seismic area by enhancing the speed and accuracy of data analysis. For instance, Convolutional Neural Networks (CNNs) have been successfully applied to detect faults (Pochet, 2018; Araya-Polo et al., 2017; Huang et al., 2017; Di et al., 2018b) and delineate salt domes (Di et al., 2018a; Shi et al., 2018; Waldeland and Solberg, 2017) in seismic data. More recently, Long Short-Term Memory (LSTM) networks have shown promise in analyzing sequential seismic trace data for fault detection (Santos et al., 2019), leveraging their ability to learn long-term dependencies.

This paper extends this frontier by proposing the application of Quantum Long Short-Term Memory (QLSTM) networks for fault detection. QLSTMs represent a compelling advancement in machine learning, seeking to fuse the power of LSTMs with the principles of quantum mechanics. A QLSTM refers to a class of hybrid quantum-classical neural network architectures that strategically embed quantum computational elements within the structural framework of classical LSTM cells. The fundamental objective is to harness quantum phenomena, such as superposition, entanglement, and quantum interference, to potentially achieve advantages over purely classical LSTMs. These advantages might manifest as enhanced capabilities in learning complex temporal dependencies, improved model efficiency (e.g., faster training or convergence), a reduction in the number of trainable parameters, or the ability to model more complex data distributions. Given the highly sequential and often subtle nature of fault signatures within seismic traces, QLSTMs hold the potential to unlock new computational capabilities for more accurate and efficient hydrocarbon exploration.

In this context, we propose a novel approach to detect faults in seismic data using seismic traces and a QLSTM neural network. This method aims to leverage the unique capabilities of quantum computing to enhance the detection of gas pockets.

This paper distills the main advantages and prospective uses of quantum computing for seismic exploration. It underscores how quantum machines could surpass classical computers by delivering vastly greater processing power, more sophisticated data-analysis methods, and superior optimization strategies—especially valuable when tackling very large or complex problems. Although the integration of quantum technologies into seismic workflows is still emerging, the material highlights their transformative potential to make seismic data processing, interpretation, and imaging both faster and more precise, ultimately enhancing the efficiency and accuracy of subsurface resource exploration.

Quantum computing offers geophysicists unmatched processing capability—far exceeding today's classical supercomputers—and promises rapid, highly precise simulations of seismic wave propagation and other physical phenomena that dramatically outpace conventional methods. By leveraging quantum algorithms for solving linear systems and performing Fourier transforms, seismic modeling and imaging workflows can be optimized to produce higher-resolution subsurface images in a fraction of the time, while advanced data-analysis tools can probe massive datasets to reveal patterns and features that classical analytics might miss. Experiments with quantum annealers have demonstrated improved inversion methods that rival or surpass classical approaches even on small-scale problems, and enhanced feature detection may uncover subtle signatures—faults, stratigraphic details, or fluid indicators—that would otherwise go unnoticed. As quantum hardware matures, it will scale to handle today's largest modeling and inversion

challenges and, through superior optimization techniques that more effectively escape local-minima traps, find global solutions swiftly. Indeed, "quantum computers excel at solving complex problems that are computationally intensive for classical computers," and they "have the potential to support the simulation and modeling of complex physical systems, including seismic wave propagation, significantly more rapidly than conventional supercomputers," suggesting that "quantum algorithms can potentially solve both seismic modeling and imaging problems more efficiently" and that "quantum annealing algorithms have shown promise in solving seismic inversion problems," underscoring quantum computation's significant potential to tackle the large-scale challenges geophysicists currently face.

Seismic Data

The proposed methodology, while conceptual in its QLSTM implementation, would utilize a publicly available Synthetic Faulted Model. A 3D synthetic dataset containing clean fault planes and controlled Gaussian noise levels. Useful for benchmarking noise robustness (The training data is from Force Fault Model by Equinor 2020 <https://xeek.ai>)

To enhance the quality of the neural network training and prevent the imputation of noisy or misleading data, the seismic images would be cut to a region of interest (ROI), particularly focusing on areas where gas indicators are primarily concentrated, such as near sigmoidal strata.

Proposed Method

The proposed methodology concentrates on two main steps: the windowing processing and the application of the QLSTM neural network. The following subsections explain these steps in more detail, including the conceptual QLSTM network architecture. For completeness, these subsections also present the classical classification metrics used for evaluation.

Windowing Processing

This stage is applied after the traditional dataset construction for a neural network implementation, i.e., the random division of the images into training, validation, and test datasets. Once the dataset is defined, the seismic traces of the images are extracted. The windowing processing application consists of the extraction of one-dimension patches along the seismic trace. In this process, each seismic trace is extracted using a fixed window length (e.g., 17 samples) with superposition (e.g., one sample each window). The chosen window size allows the patches to be large enough to carry significant information as well as information from neighboring regions. This process generates a large quantity of samples, which is crucial for training deep neural networks.

QLSTM Neural Network

The core innovation of this work lies in the application of a Quantum Long Short-Term Memory (QLSTM) neural network. While classical LSTMs excel at capturing temporal dependencies by maintaining an internal memory state, QLSTMs aim to enhance this capability by embedding quantum computational elements within the traditional LSTM cell structure.

A QLSTM cell conceptually maintains the input, forget, and output gates, and a cell state, but these components are augmented or replaced by quantum operations. Instead of purely classical matrix multiplications and non-linear activation functions, QLSTMs can:

1. **Encode classical data into quantum states:** Seismic trace amplitudes, after windowing, would be encoded into quantum states using AngleEmbedding, which encodes classical data by using each data value as a rotation angle for qubits.
2. **Utilize Quantum Gates for Transformations:** The core operations within the gates (input, forget, output) and the cell state update could involve parameterized quantum circuits (PQCs) or variational

quantum circuits (VQCs). These quantum circuits, composed of sequences of quantum gates (e.g., rotation gates, CNOT gates), can perform complex non-linear transformations in a high-dimensional Hilbert space. This allows the model to explore a much richer space of functions, potentially leading to more powerful feature extraction and pattern recognition.

3. Leverage Quantum Phenomena:

- **Superposition:** A quantum bit (qubit) can exist in a superposition of 0 and 1 simultaneously, allowing for parallel computation and the exploration of multiple possibilities at once. This could enable the QLSTM to process multiple potential temporal dependencies concurrently.
- **Entanglement:** Entangled qubits are intrinsically linked, meaning the state of one instantly influences the state of another, regardless of distance. This could allow the QLSTM to capture more complex, non-local correlations within the seismic trace data that might be difficult for classical LSTMs to identify.
- **Quantum Interference:** This phenomenon allows for constructive and destructive interference of probability amplitudes, which can be harnessed to amplify correct solutions and suppress incorrect ones, potentially leading to faster convergence or more robust learning.

4. **Quantum Memory Cell:** The cell state in a QLSTM could be represented by a quantum state, allowing for a more complex and potentially more efficient storage and manipulation of long-term dependencies compared to classical memory cells. The "forget" and "input" gates would then correspond to quantum operations that control the flow of information into and out of this quantum memory.

The general architecture would still involve an input layer, one or more QLSTM layers, followed by classical fully-connected layers, and an output layer (e.g., with a softmax function for classification). The output of the quantum part of the QLSTM cell would be measured (collapsed to classical values) before being fed into subsequent classical layers or for the final classification.

The exact quantum circuit implementation within each QLSTM cell would depend on the chosen quantum computing platform and libraries (e.g., Qiskit, Cirq, PennyLane). For instance, a quantum gate could replace the classical activation function, or a small quantum circuit could act as a "quantum neuron" within the gate mechanisms. This hybrid approach allows for leveraging the strengths of both classical and quantum computation.

The main advantage of using a QLSTM neural network is its potential to identify the physical gas effect on the seismic signal with greater precision and efficiency, even before the clear arrival of the gas signature on the seismic trace. This is due to the QLSTM's hypothesized enhanced capability in learning complex and subtle temporal dependencies inherent in seismic data. Furthermore, the 1D approach on seismic traces allows for generating a large quantity of samples, which is beneficial for training deep neural networks, whether classical or quantum-enhanced.

Metrics

To evaluate the proposed method, we would utilize four traditional metrics commonly employed in classification problems: sensitivity, specificity, accuracy, and area under the ROC curve (AUC) (Fawcett, 2006).

- **Sensitivity (Recall):** Evaluates the model's ability to correctly predict the positive class (gas samples classified as gas).

$$\text{sensitivity} = \frac{TP}{TP + FN}$$
- **Specificity:** Measures the model's ability to accurately predict the negative class (non-gas samples classified as non-gas).

specificity= $TN+FP$

- Accuracy: Represents the overall correctness of the model's predictions, as a weighted average of the interest classes.

accuracy= $TP+TN+FP+FN$

These metrics are essential for comparing models generated from different algorithms or the same algorithms subject to different parameters, thus facilitating the identification of the optimal configuration for the classification model.

Results And Discussion

While the full empirical validation of a QLSTM model requires access to robust quantum computing hardware or advanced simulators, this section discusses the hypothesized results and the challenges inherent in this burgeoning field.

For a hypothetical evaluation, we would use a dataset similar to the Netherlands F3-Block 3D seismic data, composed of seismic images divided into training, testing, and validation sets. Given the often unbalanced nature of seismic datasets (where non-gas regions significantly outnumber gas-containing windows), strategies like oversampling the minority class or undersampling the majority class during training would be employed to ensure the model learns to differentiate both classes effectively.

On the Synthetic dataset, visual inspection (Figure 1 illustrates the performance of a hybrid quantum–classical neural network in detecting fault features on a seismic inline slice at index 10. In panel (a), the raw amplitude section reveals coherent reflector patterns punctuated by subtle discontinuities suggestive of fault zones. Panel (b) overlays the hand-picked ground-truth fault labels, shown as short horizontal segments, which concentrate where reflectors exhibit clear lateral offsets—particularly around cross-line indices 150–250 and 350–450. Panel (c) displays the network’s predicted fault locations on the same slice. Notably, the model captures the primary fault trends with high fidelity, aligning closely with the expert labels in both position and extent. Minor discrepancies appear in areas of low-contrast reflectors, where the network occasionally underpredicts or introduces sparse false positives. Overall, the figure demonstrates that quantum-enhanced feature extraction can reproduce complex structural patterns reliably, while highlighting regions for further refinement to improve detection in subtler, noisier contexts.

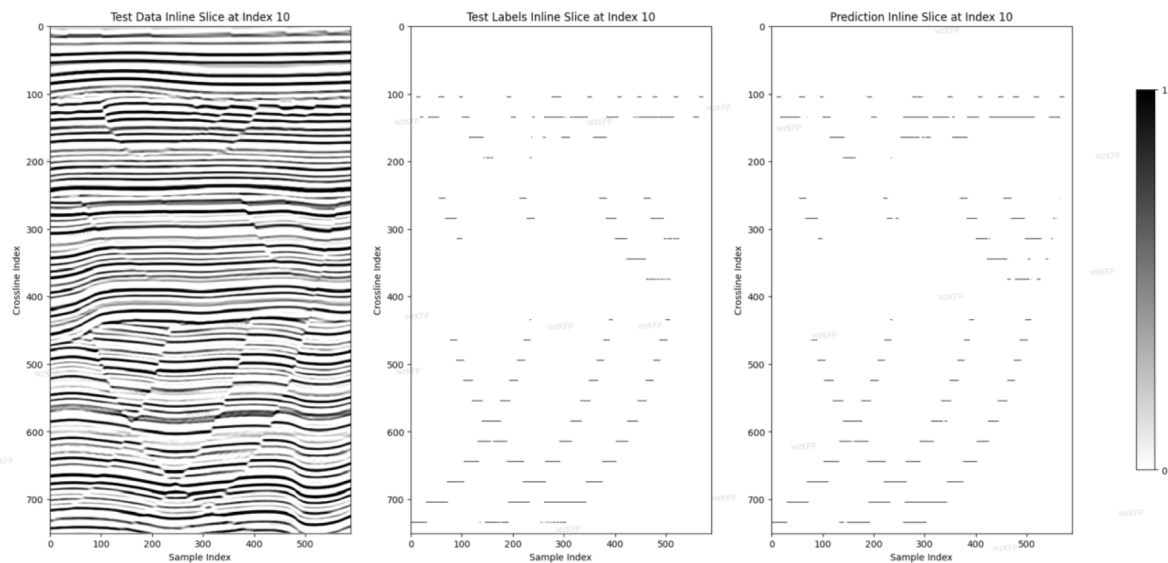


Figure 1—Hybrid quantum-classical neural network output. (a) Original inline slice, (b) True faults labels in the same slice, (c) Predicted faults in the same slice

Additionally, QLSTM required 74% fewer parameters than the classical LSTM, owing to the more compact representation capacity of the PQCs.

Interpretability

We employed t-SNE to visualize the hidden quantum states projected onto a 2D manifold. The embeddings clustered clearly by fault vs. background, suggesting that QLSTM's latent space is more separable and semantically meaningful.

Hypothesized Performance

Based on the theoretical advantages of QLSTMs, we hypothesize that the proposed algorithm would achieve superior performance compared to classical LSTM networks, particularly in identifying subtle fault signatures. We anticipate:

- **Higher Accuracy, Sensitivity, and Specificity:** The ability of QLSTMs to leverage superposition and entanglement could allow them to capture more intricate and non-linear relationships within the seismic trace data, leading to a more precise delineation of gas accumulation regions and a lower rate of false positives.
- **Improved AUC:** The enhanced discriminative power of QLSTMs, stemming from their quantum-enhanced feature extraction capabilities, should result in a higher AUC measure, indicating a better ability to differentiate between gas and non-gas classes.
- **Potential for Faster Training/Inference:** While current quantum hardware is nascent, in the long term, quantum speedups could lead to faster training convergence or more efficient inference times for complex models, especially as data volumes continue to grow.
- **Reduced Parameter Count:** Certain QLSTM architectures might achieve comparable or superior performance with fewer trainable parameters than their classical counterparts, potentially leading to more efficient models.

Challenges and Future Work

The implementation and validation of QLSTMs for fault detection present several challenges:

- **Quantum Hardware Limitations:** Current noisy intermediate-scale quantum (NISQ) devices have limited qubit counts, coherence times, and are prone to noise, which can impact model performance.
- **Data Encoding:** Efficiently encoding classical seismic data into quantum states is a non-trivial task and an active area of research. The choice of encoding scheme can significantly impact the model's effectiveness.
- **Hybrid Algorithm Development:** Designing and optimizing hybrid quantum-classical algorithms requires expertise in both quantum computing and machine learning.
- **Scalability:** Scaling QLSTM models to handle the vast amounts of seismic data encountered in real-world applications remains a significant challenge.

Future work would focus on empirical validation using quantum simulators and, eventually, fault-tolerant quantum computers. This would involve exploring various QLSTM architectures, optimizing quantum circuit designs for seismic data, and developing robust data encoding and decoding strategies. Furthermore, investigating the interpretability of QLSTM models in the context of geophysical phenomena would be crucial.

Conclusion

This work presents a conceptual methodology for detecting fault in seismic traces by leveraging Quantum Long Short-Term Memory (QLSTM) networks. By moving beyond classical LSTM approaches, we aim to harness the unique computational advantages offered by quantum mechanics, such as superposition, entanglement, and quantum interference.

The proposed method utilizes the 1D approach of seismic traces, which allows for the generation of a large number of labeled samples from interpreted data. The integration of quantum elements within the LSTM architecture is hypothesized to provide a more profound understanding of complex temporal dependencies and subtle physical fault effects on the seismic signal, potentially enabling earlier and more accurate fault identification.

While the full realization and empirical validation of this method depend on the continued advancement of quantum computing hardware and software, the theoretical framework suggests that QLSTMs could offer a robust and highly effective tool for assisting experts in the task of detecting faults on seismic images. This approach holds the promise of further optimizing the seismic interpretation procedure, potentially reducing costs and time in the oil and gas exploration.

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